



Industrial Artificial Intelligence for industry 4.0-based manufacturing systems

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ABSTRACT

The recent White House report on Artificial Intelligence (AI) (Lee, 2016) highlights the significance of AI and the necessity of a clear roadmap and strategic investment in this area. As AI emerges from science fiction to become the frontier of world-changing technologies, there is an urgent need for systematic development and implementation of AI to see its real impact in the next generation of industrial systems, namely Industry 4.0. Within the 5C architecture previously proposed in Lee et al. (2015), this paper provides an insight into the current state of AI technologies and the eco-system required to harness the power of AI in industrial applications.

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1. Introduction to industrial Artificial Intelligence

Artificial Intelligence (AI) is a cognitive science with rich research activities in the areas of image processing, natural language processing, robotics, machine learning etc. Historically, Machine Learning and AI have been perceived as black-art techniques and there is often a lack of compelling evidence to convince industry that these techniques will work repeatedly and consistently with a return on investment. At the same time, the performance of machine learning algorithms is highly dependent on a developer's experience and preferences. Hence, the success of AI in industrial applications has been limited. On the contrary, Industrial AI is a systematic discipline, which focuses on developing, validating and deploying various machine learning algorithms for industrial applications with sustainable performance. It acts as a systematic methodology and discipline to provide solutions for industrial applications and function as a bridge connecting academic research outcomes in AI to industry practitioners.

AI-driven automation has yet to have a quantitatively major impact on productivity growth [1]. Besides present day industries are facing new challenges in terms of market demand and competition. They are in need of a radical change known as Industry 4.0. Integration of AI with recent emerging technologies such as Industrial Internet of Things (IIoT) [3], big data analytics [4–6], cloud computing [7–9] and cyber physical systems [2,10–11] will enable

operation of industries in a flexible, efficient, and green way. Since Industrial AI is in infancy stage, it is essential to clearly define its structure, methodologies and challenges as a framework for its implementation in industry. To this end, we designed an Industrial AI ecosystem, which covers the essential elements in this space and provides a guideline for better understanding and implementing it. Furthermore, the enabling technologies that an Industrial AI system can be built upon are described. Fig. 1-a provides a schematic comparison of the desired system performance of Industrial AI with other learning systems over time.

2. Key elements in Industrial AI: ABCDE

The key elements in Industrial AI can be characterized by 'ABCDE'. These key elements include Analytics technology (A), Big data technology (B), Cloud or Cyber technology (C), Domain knowhow (D) and Evidence (E). Analytics is the core of AI, which can only bring value if other elements are present. Big data technology and Cloud are both essential elements, which provide the source of the information (data) and a platform for Industrial AI. While these elements are essential, domain knowledge and Evidence are also important factors that are mostly overlooked in this context. Domain knowhow is the key element from the following aspects: 1) understanding the problem and focus the power of Industrial AI into solving it; 2) understanding the system so that right data with the right quality can be collected; 3) understanding the physical meanings of the parameters and how they are associated with the physical characteristics of a system or process; and

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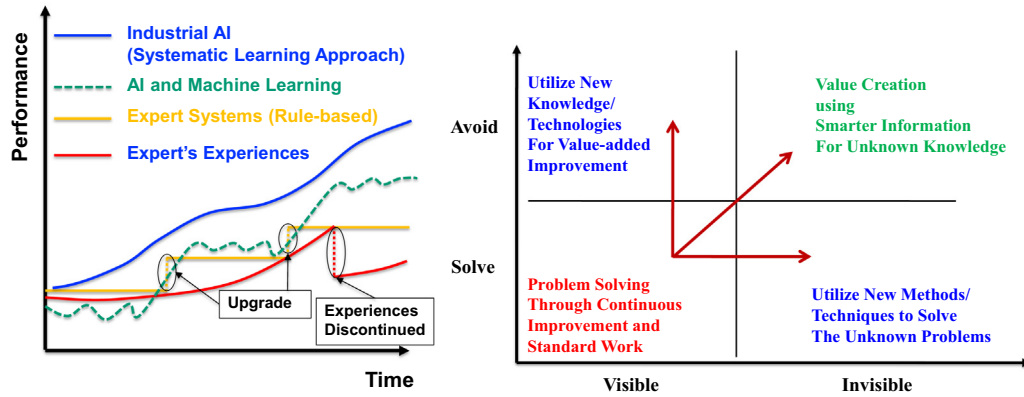


Fig. 1. a) Comparison of Industrial AI with other learning systems; b) The impact of Industrial AI: from solving visible problems to avoiding invisible.

4) understanding how these parameters vary from machine to machine. Evidence is also an essential element in validating Industrial AI models and incorporate them with cumulative learning ability. By gathering data patterns and the evidence (or label) associated with those patterns can only we improve the AI model to become more accurate, comprehensive and robust as it ages. Fig. 1-b shows how AI can drive us from visible space to invisible, and from solving the problems to avoiding them before they surface.

3. Industrial AI eco-system

Fig. 2 shows the proposed Industrial AI ecosystem, which defines a sequential thinking strategy for needs, challenges, technologies and methodologies for developing transformative AI systems for industry. Practitioners can follow this diagram as a systematic guideline for developing a strategy for Industrial AI development and deployment. Within the targeted industry, this ecosystem defines the common unmet needs such as Self-aware,

Self-compare, Self-predict, Self-optimize and Resilience. This chart also includes four main enabling technologies including Data Technology (DT), Analytic Technology (AT), Platform Technology (PT) and Operations Technology (OT). These four technologies can better be understood when put in the context of the Cyber-Physical Systems (CPS), proposed in [2]. As depicted in Fig. 3, these four technologies (DT, AT, PT and OT) are the enablers for achieving success in Connection, Conversion, Cyber, Cognition and Configuration, or 5C. This section of the paper provides a brief description of each of the mentioned technologies.

3.1. Data technologies (DT)

Data Technologies are those technologies, which enable successful acquisition of useful data with significant performance metrics across dimensions. Therefore, it becomes a co-enabler of the 'Smart Connection' step in the 5C architecture by identifying the appropriate equipment and mechanism for acquiring useful data. The other aspect of data technologies is data communication.

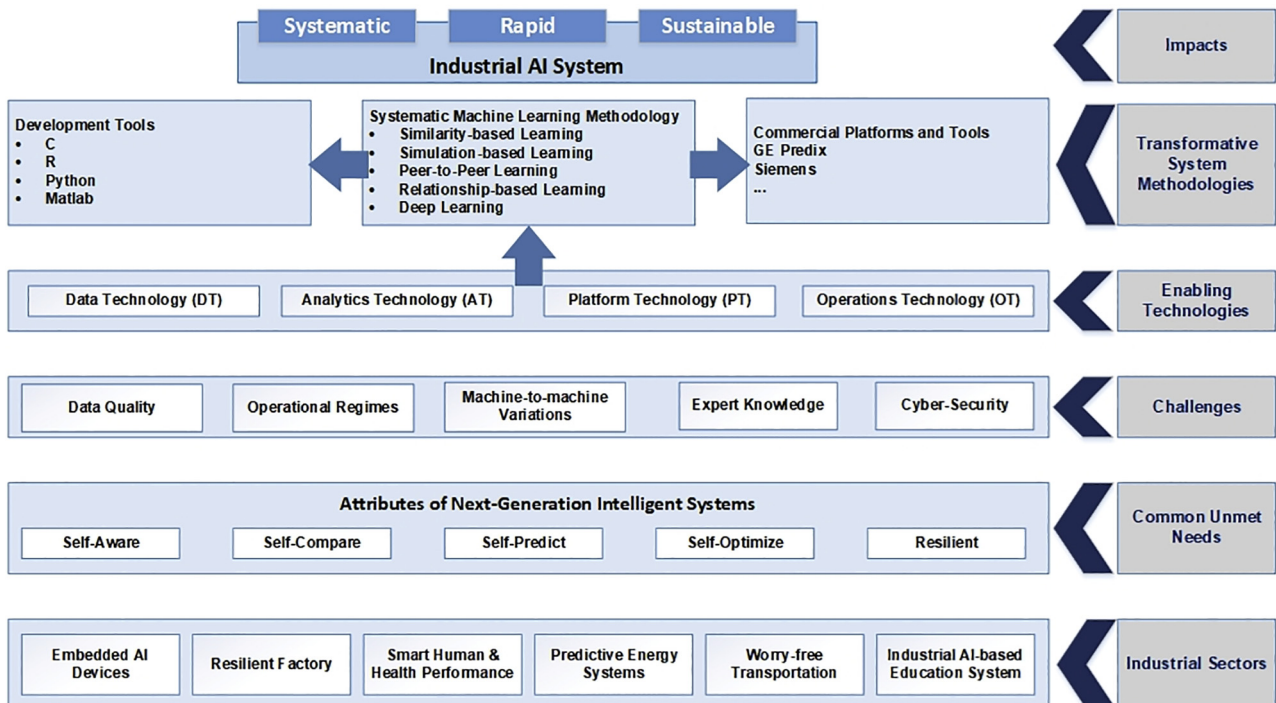


Fig. 2. Industrial AI Eco-system.

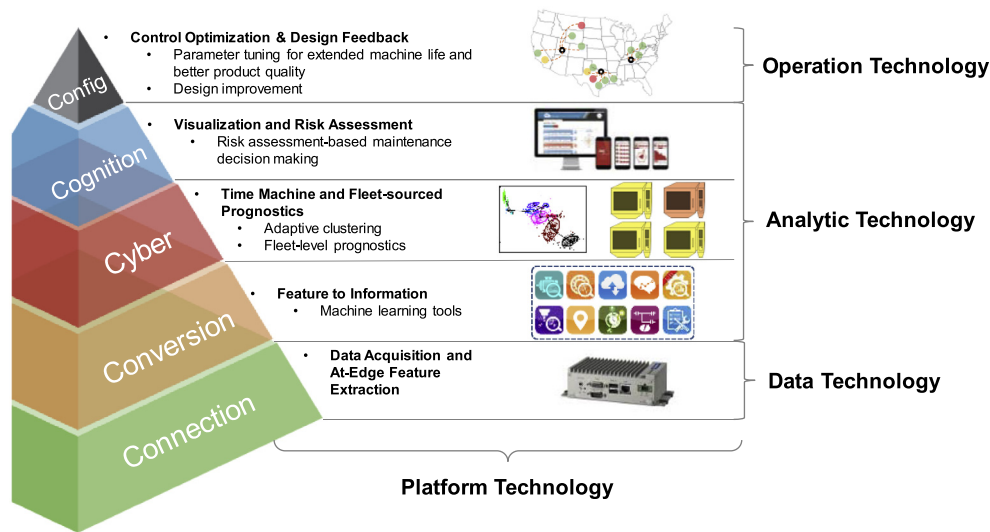


Fig. 3. Enabling technologies for realization of CPS in manufacturing.

Communication in Smart Manufacturing extends beyond the relatively straight-forward transfer of acquired data from its source to the point of analysis. It involves: 1) Interaction between manufacturing resources in the physical-space. 2) Transfer and storage of data from machines and the factory floor to the Cloud. 3) Communication from physical space to cyber space. 4) Communication from the cyber-space to the physical-space. In addition, DT needs to address 3B issues of data systems, namely, broken, bad, and background of data [6].

3.2. Analytics technologies (AT)

Analytics Technology converts the sensory data from critical components into useful information. Data driven modeling uncovers hidden patterns, unknown correlations and other useful information from manufacturing systems. This information can be used for asset health prediction, such as generating a health value or a remaining useful life value, which can be used for machine prognostics and health management. Analytic Technologies integrate this information with other technologies for improved productivity and innovation.

3.3. Platform technologies (PT)

Platform technologies include the hardware architecture for manufacturing data storage, analysis and feedback. A compatible platform architecture for analyzing data is a major deciding factor for realizing smart manufacturing characteristics such as agility, complex-event processing, and so on. Three major types of platform configurations are generally found – stand-alone, embedded and cloud. Cloud computing is a significant advancement in Information and Communication Technologies with regard to computational, storage and servitization capabilities. The cloud platform can provide rapid service deployment, high level of customization, knowledge integration, and effective visualization with high scalability.

3.4. Operations technology (OT)

Operation technology here refers to a series of decisions made and actions taken based on the information extracted from data. While delivering machine and process health information to the operators is valuable, an Industry 4.0 factory goes beyond and enables machines to communicate and make decisions based on

the provided insight. This machine-to-machine collaboration can be between two machines in a shop floor, or machines in two different factories far apart. They can share their experience on how adjusting specific parameters can optimize performance, and adjust their production based on the availability of other machines. In an industry 4.0 factory, Operations technology is the last step leading to the following four capabilities: 1) Self-aware 2) Self-predict, 3) Self-Configure and 4) Self-Compare.

4. Case Study: Intelligent spindle system

This section describes the application and implementation of the Industrial AI architecture framework described in Section 3 to machine tool spindle of a Computer numerical control (CNC) machine. In manufacturing industry, the health condition of the machine tool spindle is of major importance and this case study aims to show how Industrial AI-powered by the four enabling technologies can provide a full solution for real-time monitoring and performance prediction of a machine tool spindle. This system is designed to minimize maintenance costs and optimize product quality, simultaneously. Based upon Fig. 3, the first step in the guideline is to consider the common unmet needs in this application area. For addressing the unmet needs (a self-aware and self-optimizing machine) the challenges of 1) data quality, 2) multi-regime complexity, 3) machine-to-machine variation, 4) incorporating expert system and 5) complexity of multi-source data need to be considered. Fig. 4 provides an overview of how DT, AT, PT and OT are utilized to address these challenges and develop an intelligent spindle system.

5. Challenges of Industrial Artificial Intelligence

The expectations from Industrial AI are versatile and enormous and even a partial fulfilment of these expectations would represent unique and real challenges of applying AI to industries. Among the existing challenges and complexities, the following ones are of higher importance and priority:

5.1. Machine-to-machine interactions

While AI algorithms can accurately map a set of inputs to a set of outputs, they are also susceptible to small variations in the inputs caused by variations from machine to machine. It needs to

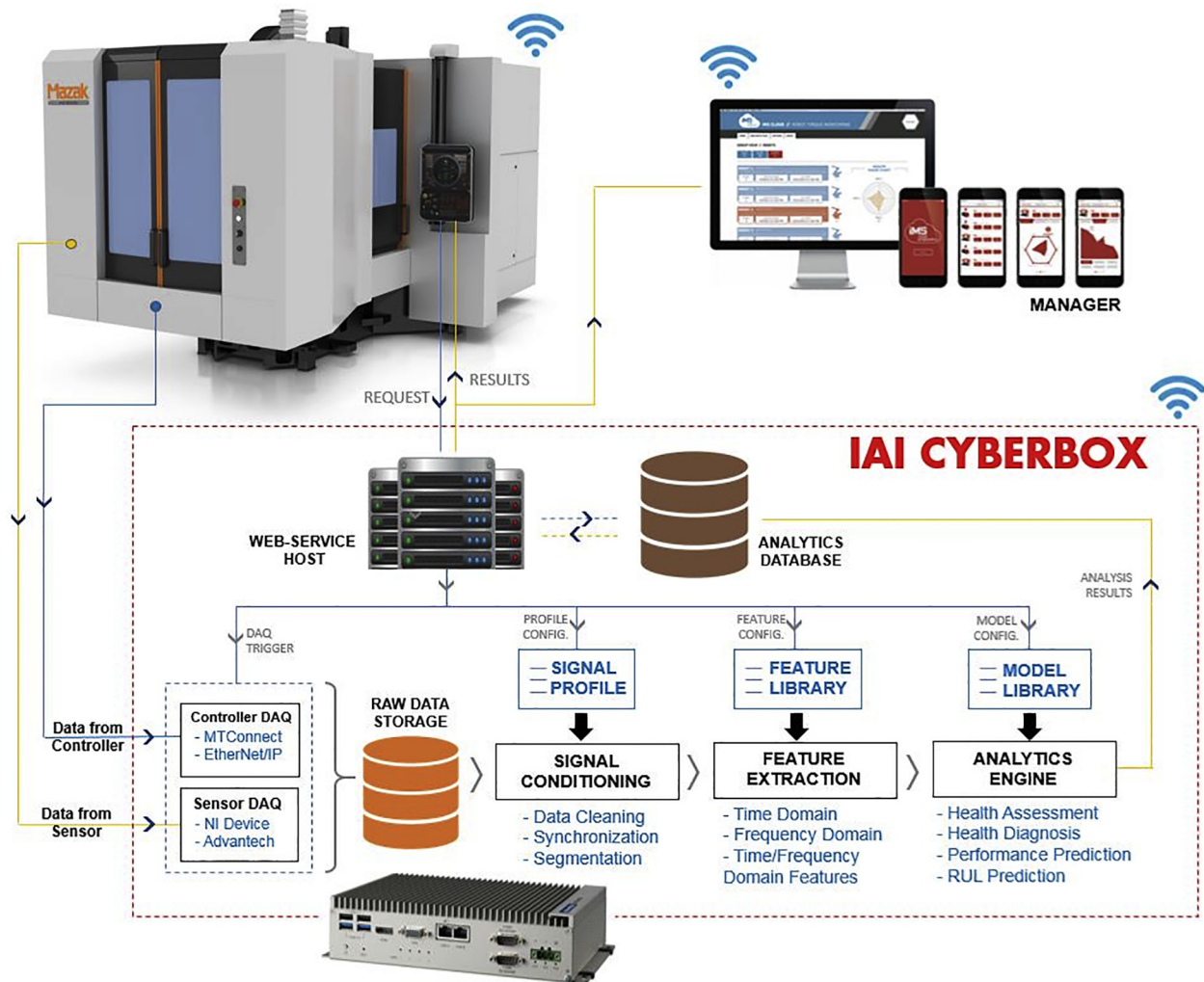


Fig. 4. Platform technology for intelligent spindle.

ensure that individual AI solutions do not interfere/conflict with the working of other systems, further down the line.

5.2. Data quality

AI algorithms require massive and clean data sets with minimum biases. By learning from inaccurate or inadequate data sets, the downstream results can be flawed.

5.3. Cybersecurity

The increasing use of connected technologies makes the smart manufacturing system vulnerable to cyber risks. Currently, the scale of this vulnerability is under-appreciated and the industry is not prepared for the security threats that exist [12].

6. Conclusion

As AI emerges from science fiction to become the frontier of world-changing technologies, there is an urgent need for systematic development and implementation of AI to see its real impact in the next generation of industrial systems, namely Industry 4.0. This study aims to define the term Industrial AI and put it into the perspective of Industry 4.0 paradigm. In addition, by providing an overview of the Industrial AI eco-system in today's manufactur-

ing, this paper aims to provide a guideline for strategizing the efforts toward realization of Industrial AI systems.

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